Table 1: The nine BDA benchmark datasets used for evaluation. All datasets are publicly available CRISPR perturbation screens. Hit rate = $n_{\text{hits}}/n_{\text{genes}}$. Batch is the number of genes selected per round (5 rounds per experiment).

Dataset	Task	Screen	n_{genes}	n_{hits}	Hit rate	Batch
IFN-γ	Cytokine regulation (Jurkat)	KO	17,785	802	4.5%	128
IL-2	Cytokine regulation (Jurkat)	KO	18,273	481	2.6%	128
Sanchez21 (up)	Tau protein upregulation (neurons)	KO	17,807	859	4.8%	128
Sanchez21 (dn)	Tau protein downregulation (neurons)	KO	17,807	871	4.9%	128
Carnevale22	T-cell fitness, adenosine pathway	KO	18,224	868	4.8%	128
Scharenberg22	Lysosomal choline recycling	KO	1,029	49	4.8%	32
Steinhart	GD2 surface expression (CAR-T)	CRISPRa	18,144	145	0.8%	128
K562-Essential	Essential genes for K562 survival	KO	623	63	10.1%	32
K562-GWPS	Genome-wide fitness screen (K562)	KO	9,193	924	10.1%	128

Table 2: Hit ratio at round 5 across all nine benchmark datasets. All methods are reported over 5 seeds. **Bold** = best per row; underline = second best. Waddington is our proposed method (C-arm), using DeLM-verified cross-experiment memory.

Dataset	Random	LOO-LightGBM	OnlineAdapt.	A: Coreset	B: LLM	C: Waddington
IFN-γ	0.029	0.168	0.183	0.100	0.160	**0.190**
IL-2	0.031	0.306	0.314	0.139	0.257	**0.347**
Sanchez21 (up)	0.037	0.077	0.087	0.031	0.063	**0.087**
Sanchez21 (dn)	0.029	0.091	**0.101**	0.078	0.071	0.099
Carnevale22	0.024	0.048	0.047	0.054	**0.059**	0.056
Scharenberg22	0.102	0.449	0.449	0.286	**0.478**	0.465
Steinhart	0.021	0.076	0.090	0.090	0.152	**0.159**
K562-Essential	0.254	0.492	0.476	0.270	**0.559**	0.556
K562-GWPS	0.065	0.247	0.273	0.156	0.225	**0.347**
Average	0.066	0.217	0.224	0.134	0.225	**0.256**

Waddington (C) achieves the best average hit ratio (0.256), outperforming the pure-LLM baseline B by 13.7% relative (+0.031) and the algorithmic baseline A by 91.6% (+0.122). It surpasses or matches every other method on 5 of 9 datasets; on the remaining four (Sanchez21-dn, Carnevale22, Scharenberg22, K562-Essential) it is a close runner-up, trailing the leader by at most 0.013, as either OnlineAdaptive or B (LLMReasoning) narrowly benefits from a strong ML signal or a narrow, well-characterised biological prior. Notably, Waddington strongly exceeds OnlineAdaptive (the PerTurboAgent-style ML-only method) on K562-GWPS (+0.075) and IFN-γ (+0.007), demonstrating the value of combining online ML with LLM reasoning.

Each component provides an additive gain: the cross-experiment LOO prior (+0.083) contributes most, followed by the LLM biological prior (+0.022 when combined with the LOO prior), and online ML retraining (+0.009). Cross-experiment memory contributes $\approx$0 in this leave-one-out benchmark (in fact marginally negative, -0.004; see ablation in Table **??**), as the LLM's parametric knowledge already subsumes the explicit memory entries.

Table 3: Stepwise improvement in average hit ratio at round 5 (9 datasets, 5 seeds). Each row introduces one additional component over the previous configuration.

Configuration	Key addition	avg hit@R5	Δ
Random	—	0.066	—
Coreset (A)	Diversity-maximising selection	0.134	+0.068
LOO-LightGBM (static)	Cross-expt. ML prior	0.217	+0.083
OnlineAdaptive	In-expt. ML retraining	0.224	+0.007
LLM Reasoning (B)	LLM biological prior (no ML)	0.225	(*parallel branch*)
C − ML (static LOO + LLM)	Combine LOO prior with LLM	0.247	+0.022 vs. B
Waddington (C, ours)	Add online ML retraining	**0.256**	+0.009

Table 4: Ablation study: hit ratio at round 5 (5 seeds, mean). C is the full Waddington arm. **C − Mem**: cross-experiment memory removed from the LLM prompt. **C − LLM**: LLM removed; online ML only. **C − ML**: ML frozen at LOO prior; no in-experiment retraining. **Shuffled**: all gene names shown to the LLM replaced with anonymous identifiers (`GENE_00001`, . . .). **Bold** = best per row; red = largest drop from C.

Dataset	C	C − Mem	C − LLM	C − ML	Shuffled	Δ_{min}
IFN-γ	0.190	**0.191**	0.190	0.178	0.175	−0.015
IL-2	0.347	0.341	0.349	**0.355**	0.351	−0.007
Sanchez21 (up)	0.087	0.092	**0.098**	0.077	0.080	−0.011
Sanchez21 (dn)	**0.099**	0.096	0.093	0.094	0.092	−0.007
Carnevale22	0.056	0.055	0.051	**0.063**	0.059	−0.005
Scharenberg22	0.465	0.482	**0.592**	0.469	0.551	+0.004
Steinhart	**0.159**	0.138	0.090	0.156	0.076	−0.083
K562-Essential	0.556	**0.603**	0.476	0.524	0.524	−0.079
K562-GWPS	**0.347**	0.343	0.343	0.309	0.343	−0.038
Average	0.256	**0.260**	0.253	0.247	0.250	

Memory ($\Delta = +0.004$). Removing the cross-experiment memory prompt has no measurable benefit; the average in fact rises slightly (+0.004) without it. LLM parametric knowledge already captures the biological context provided by memory entries in this leave-one-out benchmark, making the explicit memory module redundant. DeLM-style verified admission (a mechanical strategy-label check plus an LLM verifier that rejects hallucinated gene names) keeps the retained entries factually grounded, but does not change this conclusion: the marginal value of memory in the LOO setting remains $\approx$0.

LLM ($\Delta = -0.003$ overall; up to -0.079 per dataset). The LLM component is critical for pathway-specific tasks: Steinhart (−0.069, −44%) and K562-Essential (−0.079, −14%), where molecular pathway knowledge encoded in the LLM (e.g., GD2 biosynthesis genes, core essential gene families) cannot be recovered from PPI or DepMap features alone. For Scharenberg22, however, removing the LLM *improves* performance (+0.127), as the LLM's biological prior conflicts with the strong ML signal (leave-one-out LOO AUC = 0.825 on K562 DepMap features).

Online ML retraining ($\Delta = -0.009$). Freezing the ML model at the LOO prior produces the most consistent degradation across datasets (-0.009 average, -3.5% relative), with the largest losses on K562-GWPS (-0.038) and K562-Essential (-0.032). Online adaptation is therefore the primary source of Waddington's advantage over static ML methods.

Gene-name semantics ($\Delta = -0.006$ overall; -0.083 on Steinhart). Replacing gene names with anonymous identifiers collapses performance on pathway-specific tasks (Steinhart -0.083, -52%; K562-Essential -0.032, -6%), confirming that the LLM's contribution on those tasks is mediated by biological gene-name knowledge rather than by task description or experimental feedback. Scharenberg22 again improves ($+0.086$) when gene-name semantics are removed, mirroring the $\mathrm{C} - \mathrm{LLM}$ finding.

Figure Prompts

Figure 1 — Round Progression Curves. *Suggested prompt:* Plot a 3×3 grid of line charts, one panel per dataset. Each panel shows hit ratio on the y-axis vs. round index (1–5) on the x-axis. Draw three lines per panel: A (Coreset, dashed grey), B (LLM Reasoning, dashed orange), C (Waddington, solid blue). Add a shaded confidence band (± 1 standard error across 5 seeds). Highlight panels where $\mathrm{C} \gg \mathrm{B}$ (K562-GWPS, IL-2, IFN-γ) and panels where $\mathrm{B} \approx \mathrm{C}$ (Steinhart, Scharenberg22) with a light background tint. Title each panel with dataset name and final-round gap (C−B). Data: use per-round hit_ratio values from the `v8_results.json` output of `run_sequential.py --arms waddington_v14 llm_reasoning coreset --seeds 5`.

Figure 2 — Ablation Heatmap. *Suggested prompt:* Create a heatmap with datasets on the y-axis (9 rows) and ablation variants on the x-axis (C−Mem, C−LLM, C−ML, Shuffled, 4 columns). Cell value = hit ratio delta relative to full C (Δ = ablation $- C$). Use a diverging colormap (blue = gain, red = loss), symmetric around 0, clipped at $[-0.15, +0.15]$. Annotate each cell with the Δ value (3 decimal places). Add a bottom row showing column averages. This figure highlights: (i) the LLM is critical for Steinhart and K562-Essential; (ii) the LLM hurts Scharenberg22; (iii) online ML retraining (C−ML) consistently degrades most datasets; (iv) gene-name semantics (Shuffled) selectively affect pathway tasks.

Figure 3 — Component Decomposition Bar Chart (optional). *Suggested prompt:* Horizontal grouped bar chart. Each dataset is one group; bars represent average hit@R5 for: Random, Coreset (A), LLM (B), OnlineAdaptive, Waddington (C). Sort datasets by C−B gap (ascending). Annotate the Waddington bar with the dataset's routing category (ml_heavy / baseline / two_stage) to visualise how routing strategy correlates with the ML vs. LLM trade-off.